



CITEAM Workshop 2026

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2519465



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Section 0.1.1 Introduction to TEM/STEM



FIGURE 1.1. The electron microscope built by Ruska (in the lab coat) and Knoll, in Berlin in the early 1930s.

Transmission Electron Microscopy, A textbook for Materials Science, David B Williams and C Barry Carter, Springer, 2nd edition

2009

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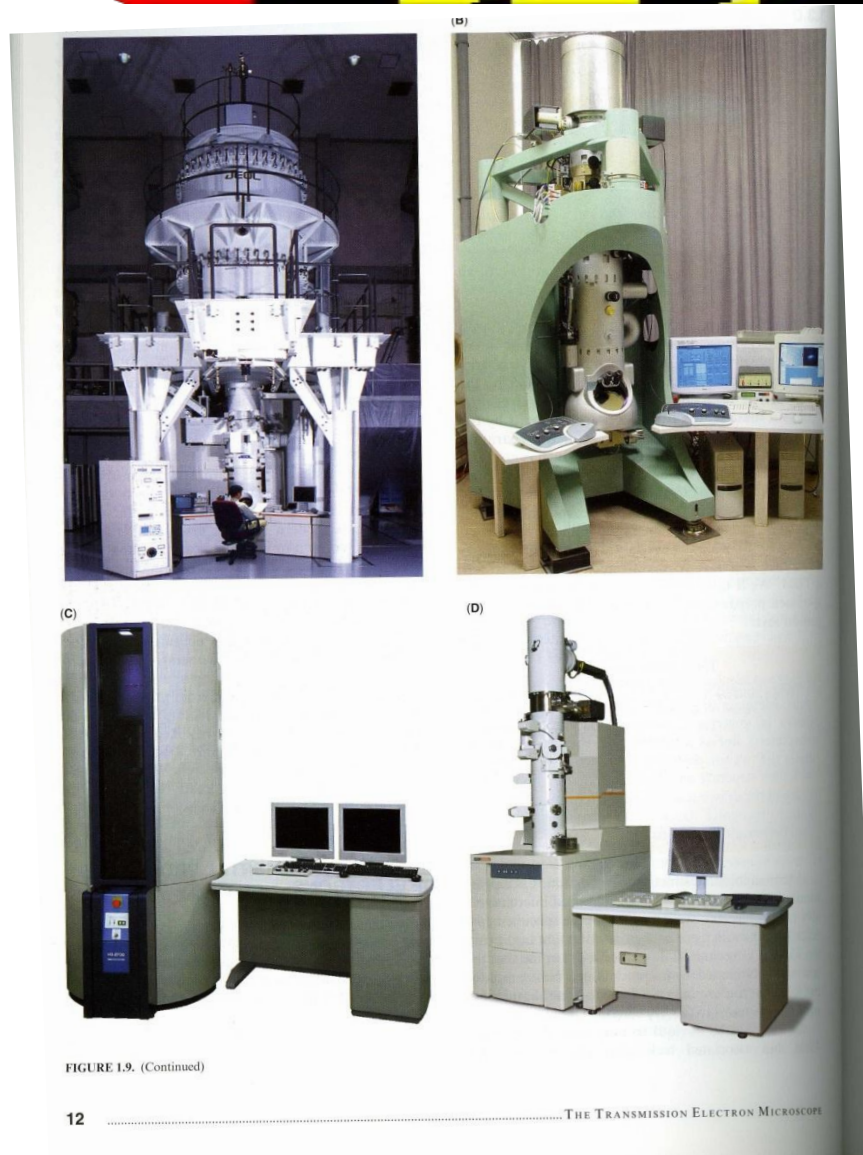


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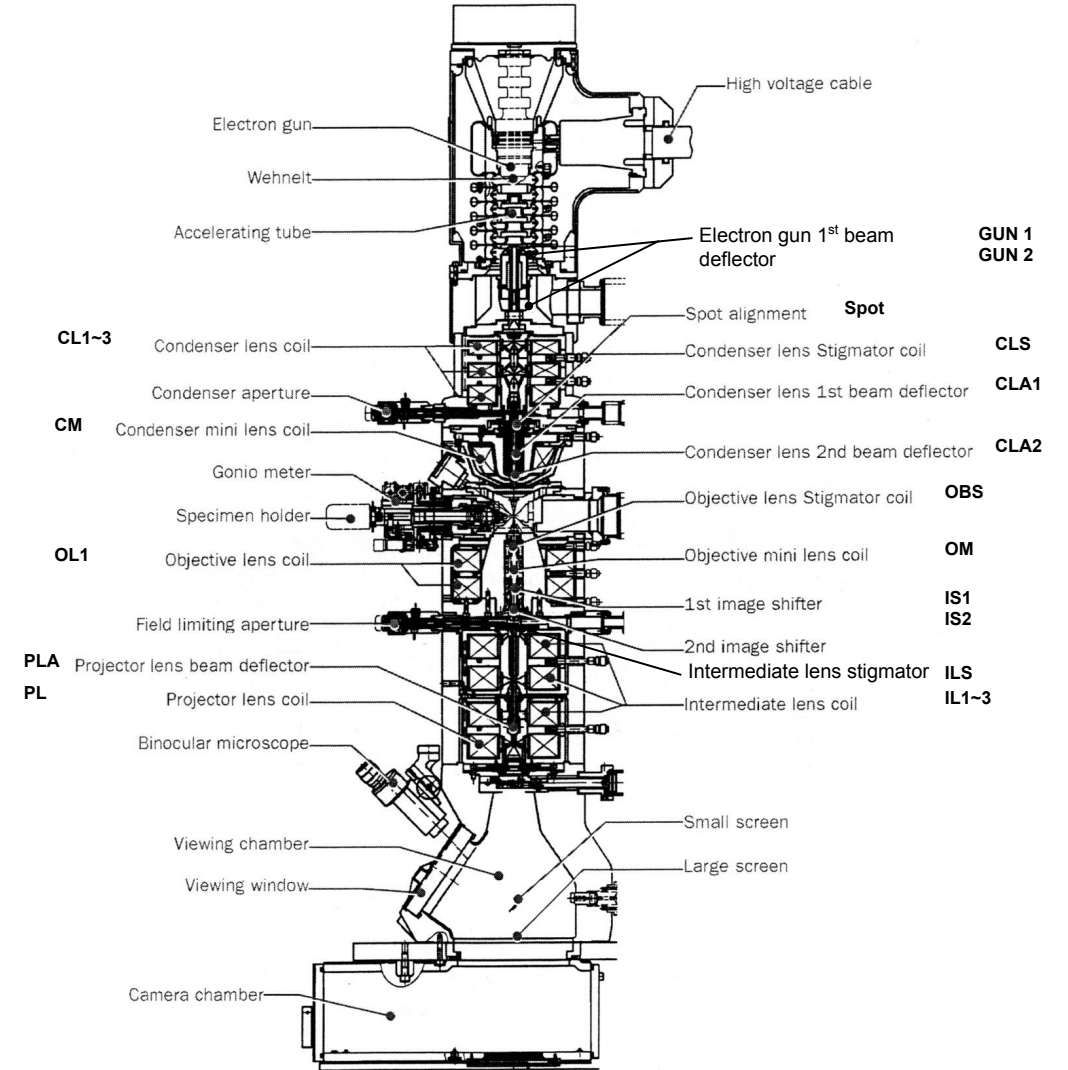


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Evolution of TEMs



Cross section of the JEOL 2100F TEM



NEOARM Highlights

NEOARM 200 F Cs corrected TEM in the IDEA Factory

- High Brightness Cold FEG
- 30-200kV
- HR Pole Piece for maximum flexibility between analytical and resolution optimization
- IDES Technology integration
- Dual, large-area SDDs for efficient EDS collection
- Highly stable probe current
- COSMO Automated Corrector alignments



Cross section of NEOARM Column

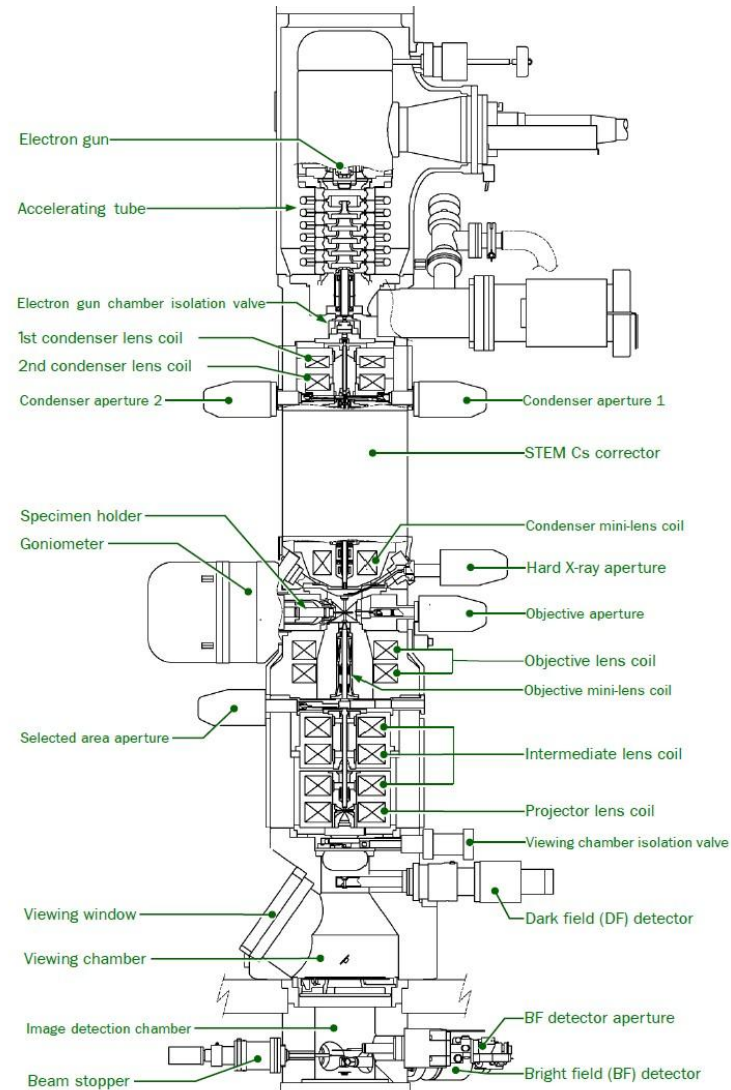


Figure 3.3 Cross section of column

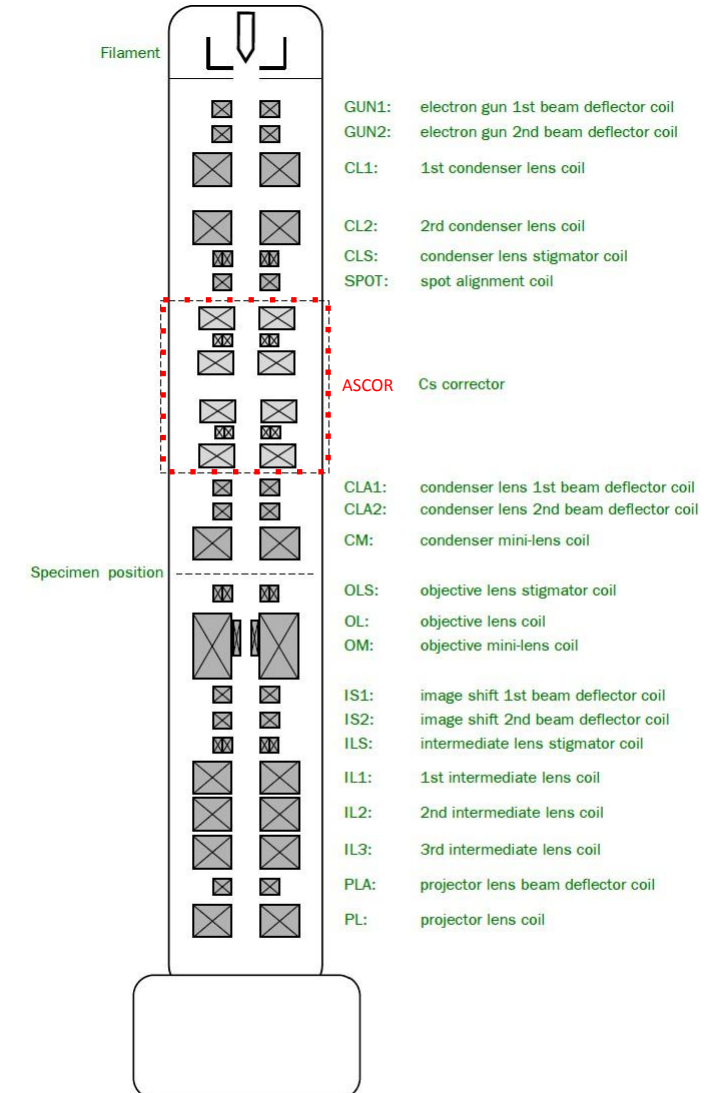
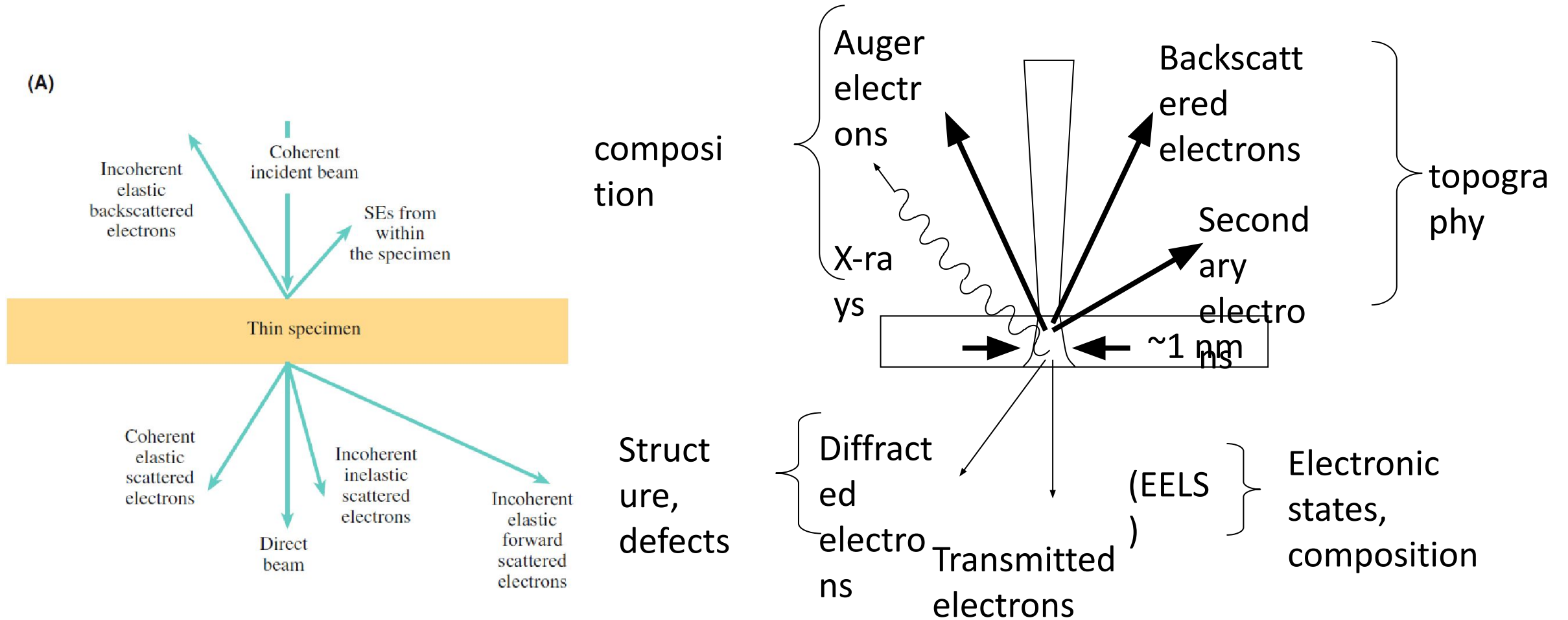


Figure 3.4 Location of coils and lenses

Electron beam sample interaction



Transmission Electron Microscopy, A textbook for
Materials Science, David B Williams and C Barry Carter,
Springer, 2nd edition 2009

Imaging and Diffraction in TEM

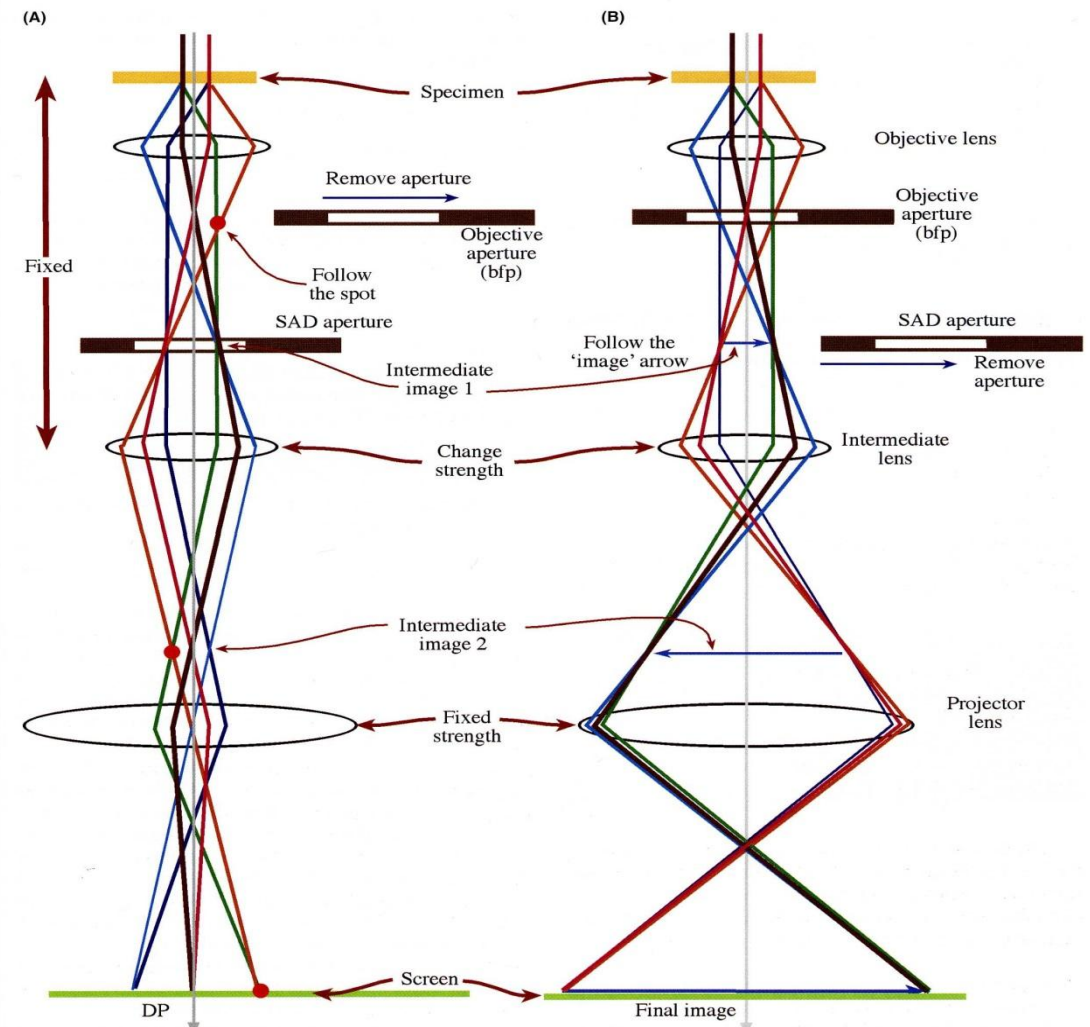
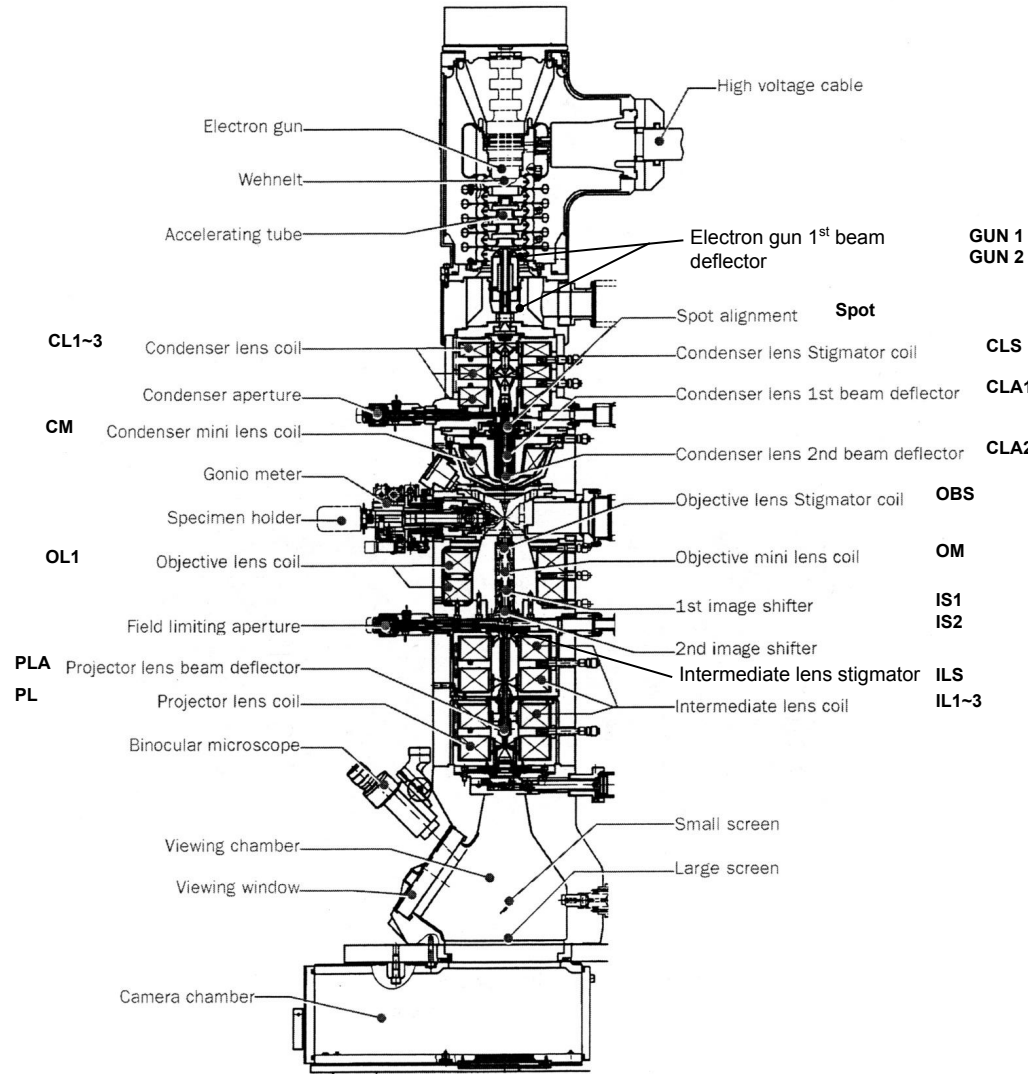
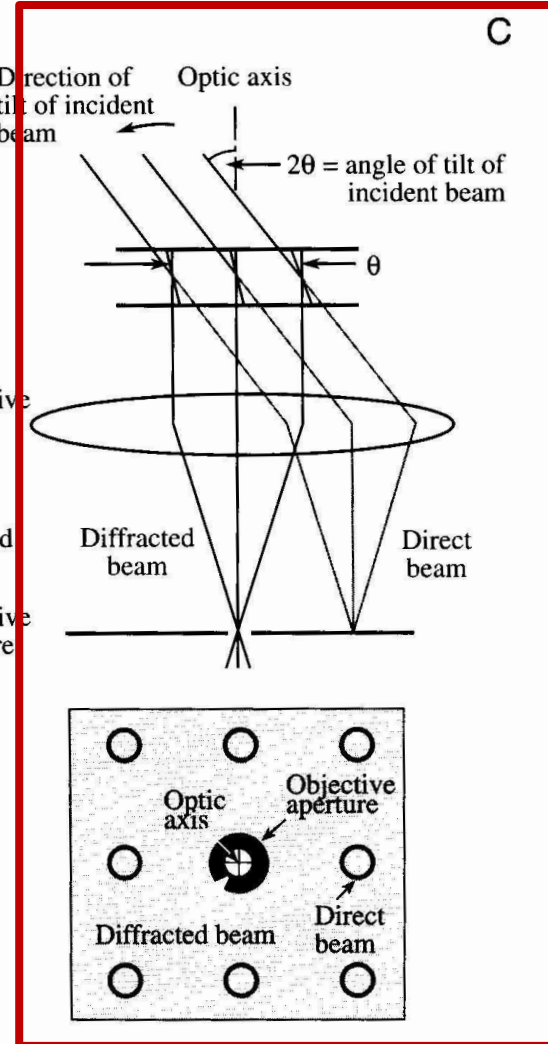
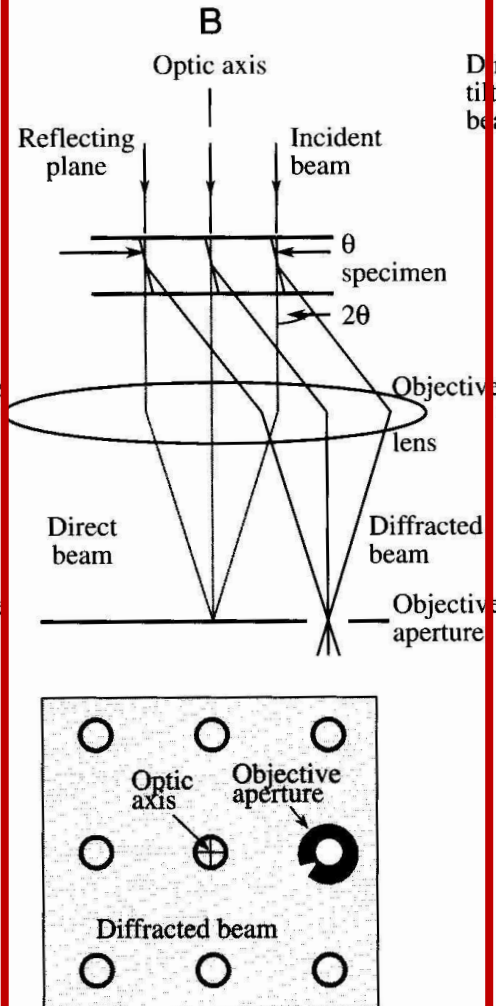
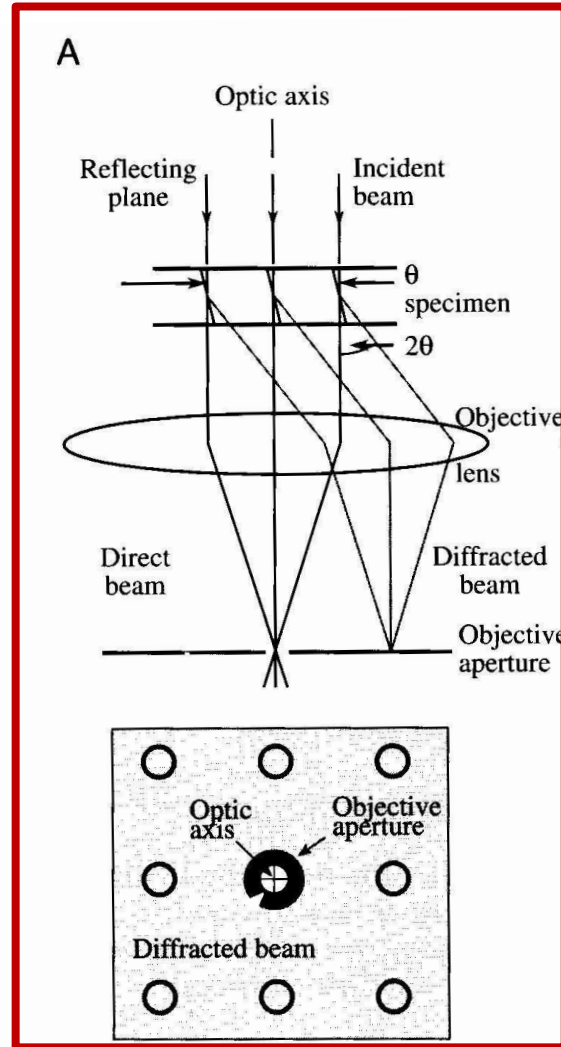


FIGURE 9.12. The two basic operations of the TEM imaging system involve (A) diffraction mode: projecting the DP onto the viewing screen and (B) image mode: projecting the image onto the screen. In each case the intermediate lens selects either the BFP (A) or the image plane (B) of the objective lens as its object. The imaging systems shown here are highly simplified. Most TEMs have many more imaging lenses, which give greater flexibility in terms of magnification and focusing range for both images and DPs. The SAD and objective diaphragms are also shown appropriately inserted or retracted. NOTE: This is a highly simplified diagram showing only three lenses. Modern TEM columns have many more lenses in their imaging systems.

Bright Field and Dark Field in TEM

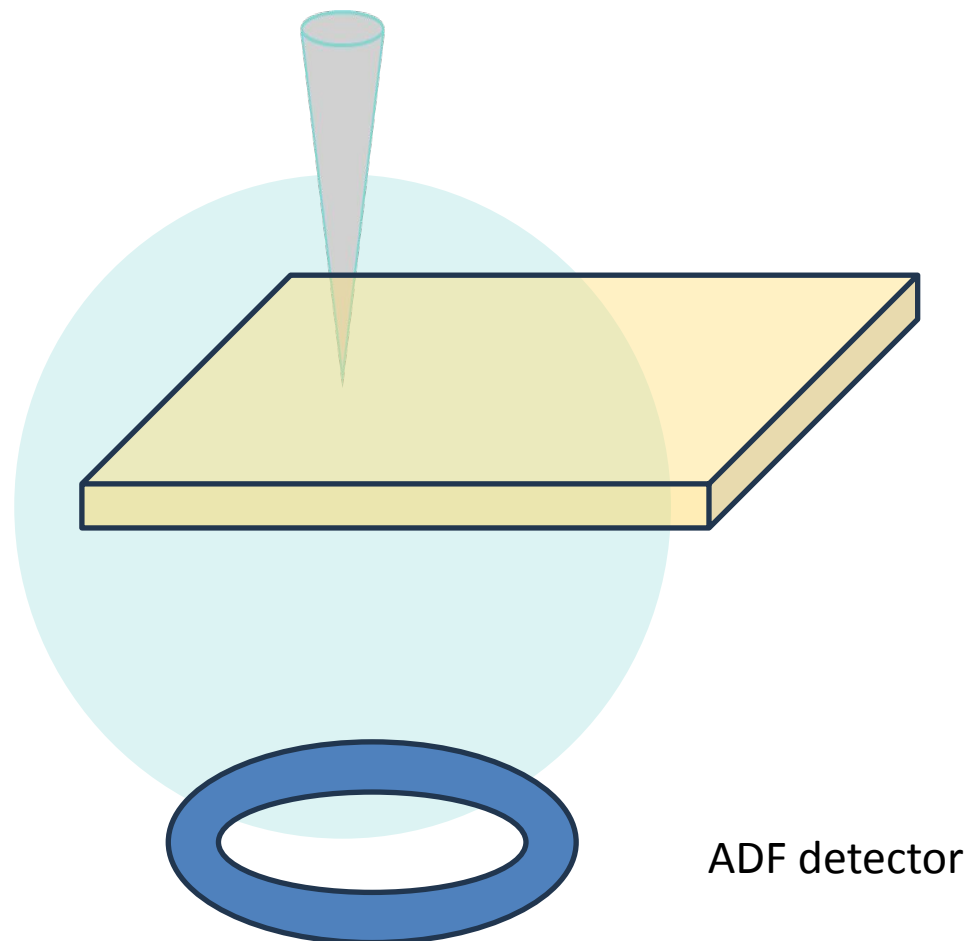
Bright Field



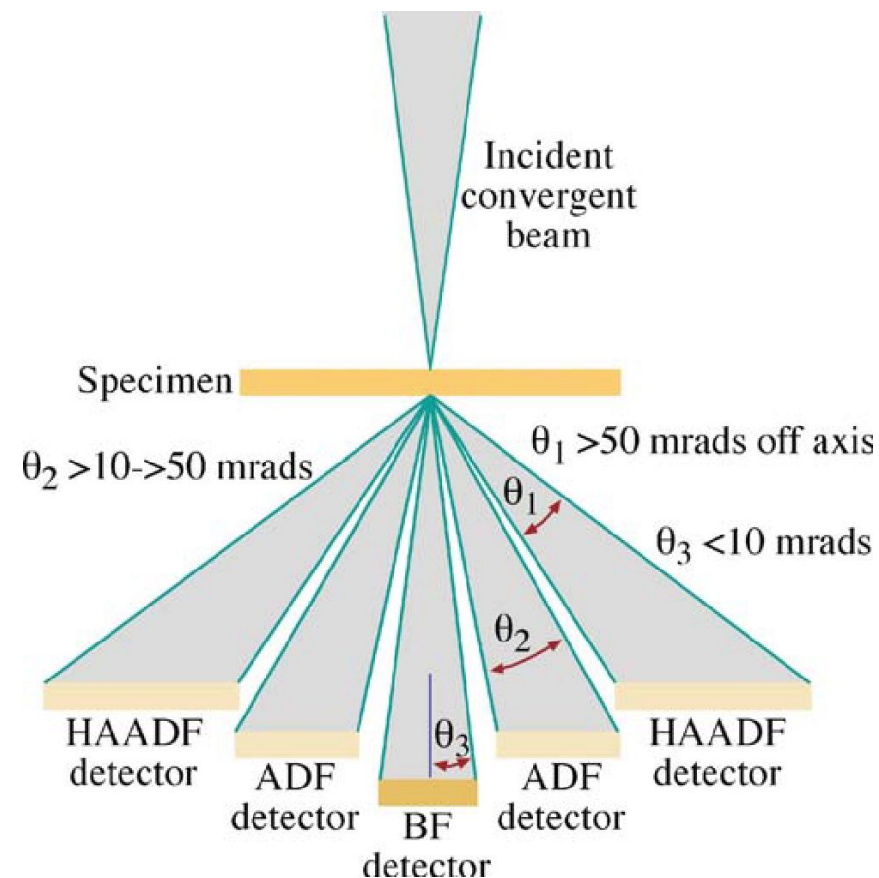
Dark Field

Figure 9.14. Ray diagrams showing how the objective lens/aperture are used in combination to produce (A) a BF image formed from the direct beam, (B) a displaced-aperture DF image formed with a specific off-axis scattered beam, and (C) a CDF image where the incident beam is tilted so that the scattered beam remains on axis. The area selected by the objective aperture, as seen on the viewing screen, is shown below each ray diagram.

STEM imaging



In STEM mode the electron beam is scanned on an area of the sample and the selected detector collects the signal from each point



BF Bright Field detector

ADF Annular Dark Field detector

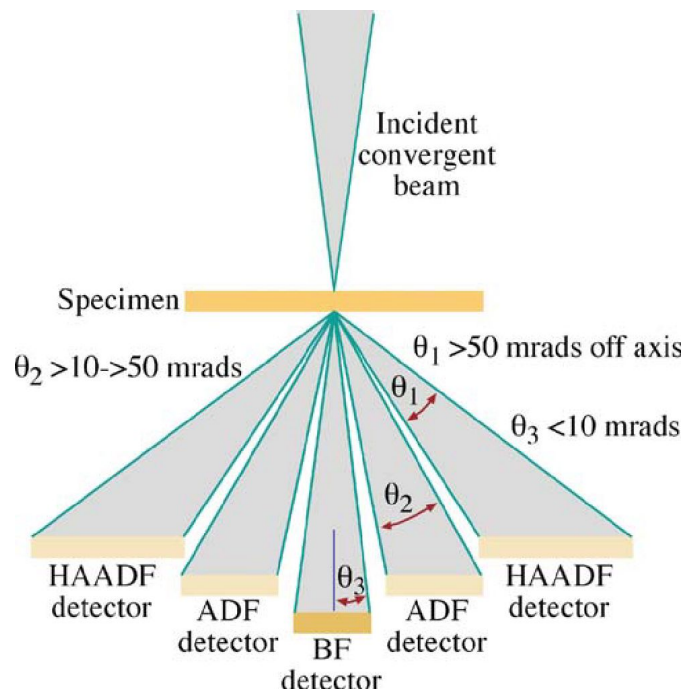
HAADF High Angle Annular Dark Field detector

Transmission Electron Microscopy, David Williams and Barry Carter, 2nded. (2009)

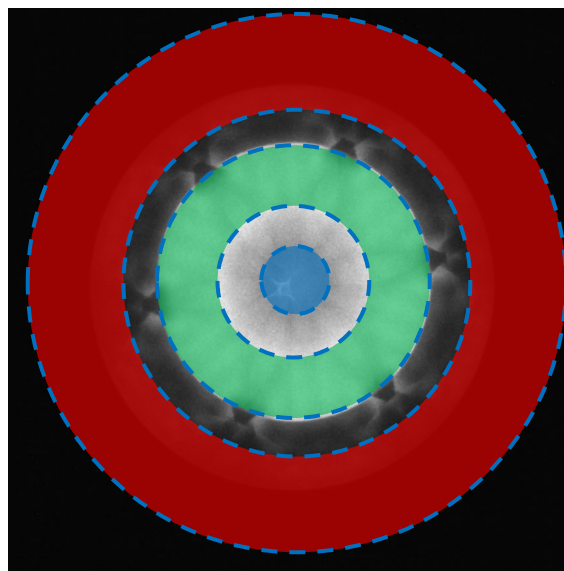
NEOARM Specifics for STEM Imaging

STEM Imaging Modes

- High-angle annular dark field (HAADF)
- Bright field (BF)
- Annular bright field (ABF)



Diffraction Pattern (STEM)



HAADF:

Z-contrast
Robust, easy to understand

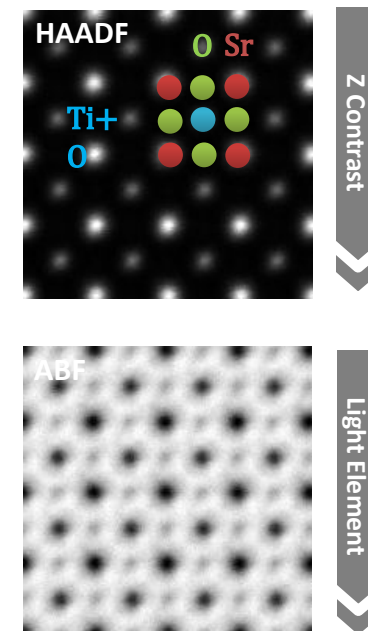
BF:

Close to conventional TEM
Phase Contrast (not straightforward)

ABF:

Enhanced contrast for light elements
Robust, easy to understand

SrTiO₃ Example



Transmission Electron Microscopy, David Williams and Barry Carter, 2nded. (2009)

From JEOL

NEOARM Specifics

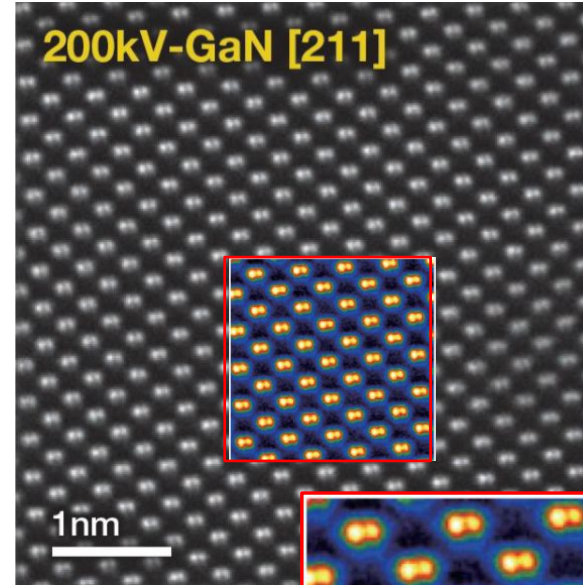
NEOARM

- High-brightness Cold Field-Emission Gun
- Column and Corrector alignments provided at 30 kV, 80 kV, 200 kV
- Resolution of .23nm in TEM
- +/- 35/30 degrees of tilt in X/Y

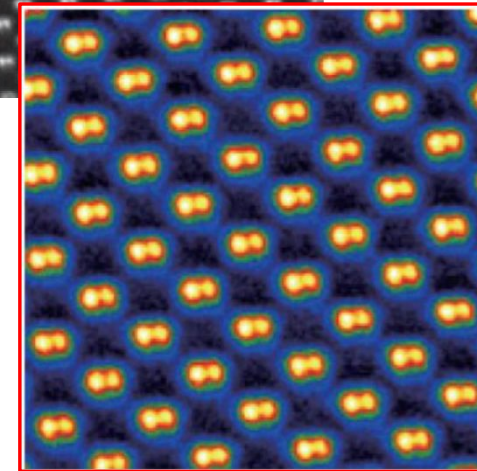
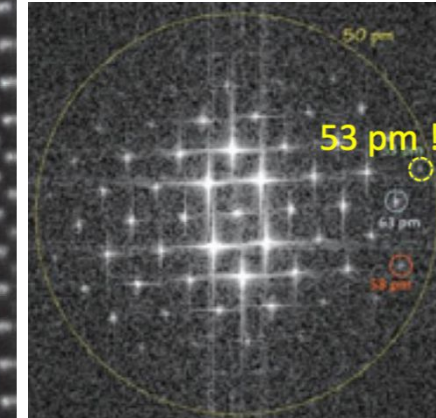
STEM C_s -Corrector

- CEOS ASCOR – Second-generation, higher-order corrector
- Enables high-resolution imaging with large convergence angles and high probe currents
- JEOL COSMO automated Corrector alignment routine

HAADF STEM Image



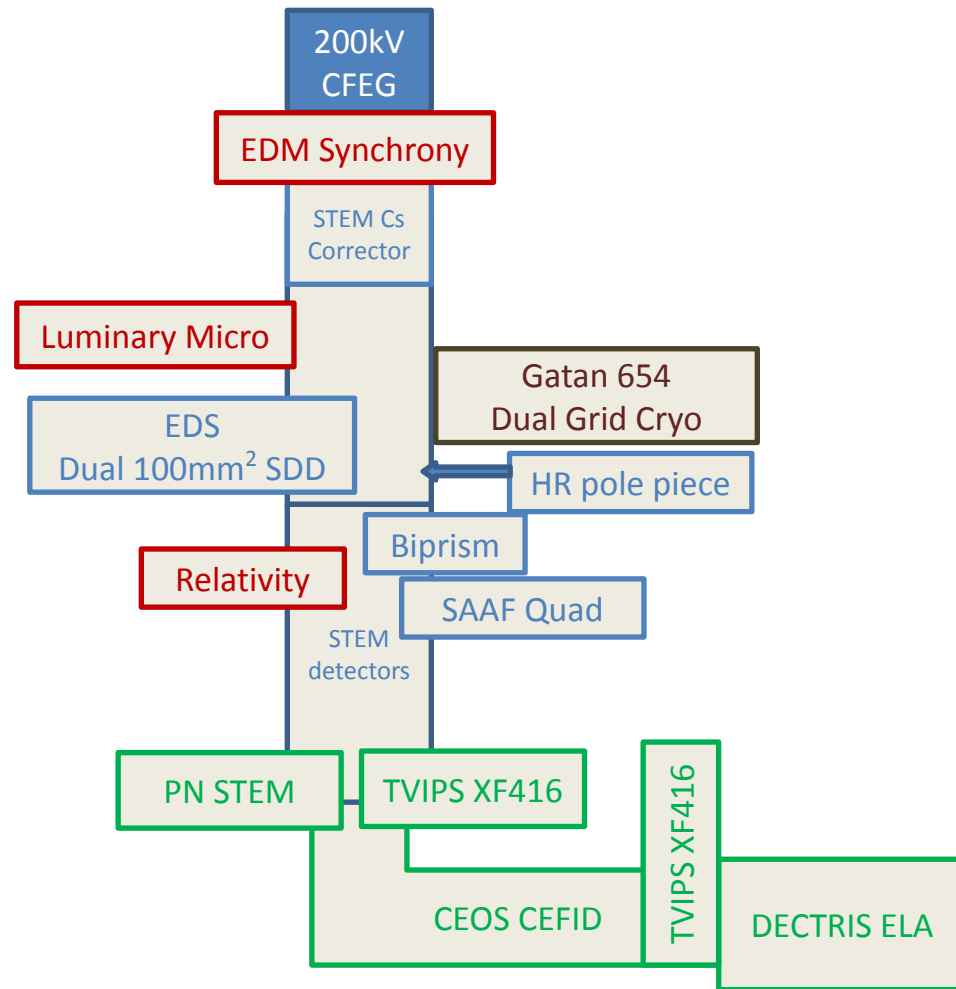
FFT



Ga-Ga dumbbells
split by 63 pm

From JEOL

JEOL NEOARM + CEFID Configuration



From JEOL

JEOL

- STEM C_s -Corrected --Aligned at 30kV, 80kV, 200kV
- Dual EDS
- STEM HAADF, STEM ABF detectors
- Segmented STEM (SAAF) Quad Detector
- STEM Lorentz & NBD alignments
- Biprism
- PyJEM (Python-based) scripting control of JEOL + IDES

IDES

- EDM Synchrony (Electrostatic Dose Modulator)
- Luminary Micro
- Relativity Sub-framing

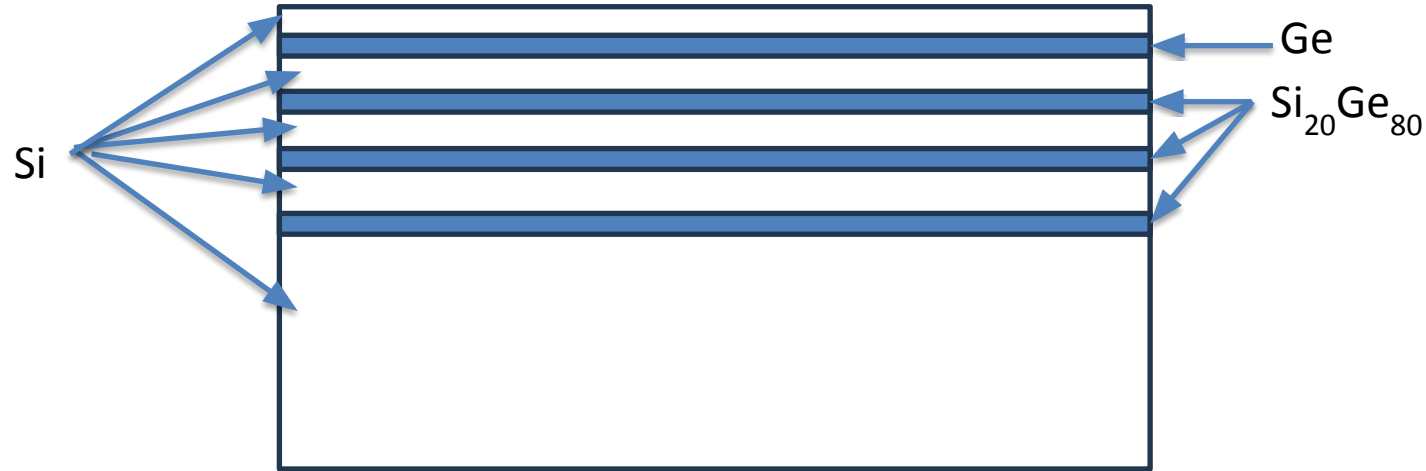
CEOS/TVIPS/DECTRIS/PN Detector

- CEFID energy filter
 - ELA detector
 - TVIPS XF416 detector
- TVIPS XF416 Bottom mount
 - Use with IDES Relativity
- PN Detector for STEM

Other

- Gatan Straining Holder (654)
- Simple Origin Dual Grid Cryo
- Protochips Poseidon holder
- Protochips Atmosphere holder
- Protochips Fusion holder

Sample 1 to be Analyzed



Sample from Riss Card at UMD

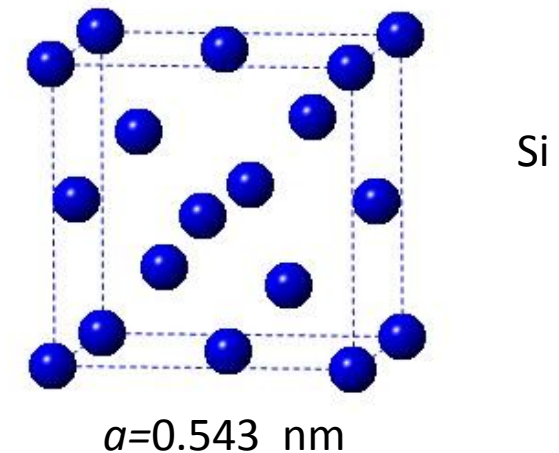
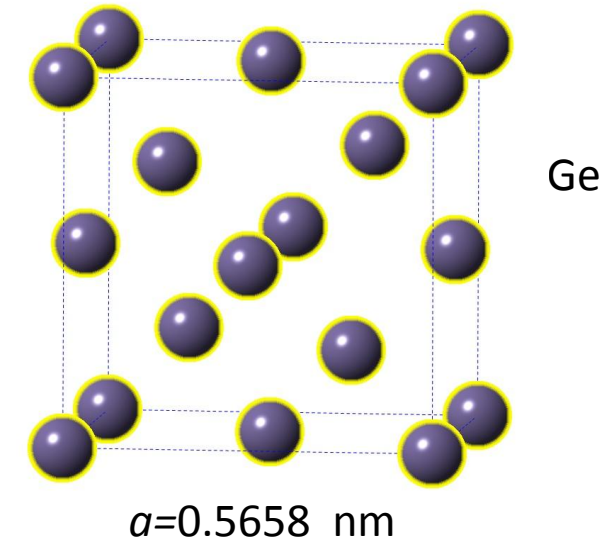
Ge and Si are in group IV of the periodic table. However, Ge is heavier than Si and has larger lattice constant. The contrast between the Si_{1-x}Ge_x layers and pure Si layers depends on x. ADF in STEM gives higher contrast because of the larger atomic mass of Ge compared to Si.

What to look for:

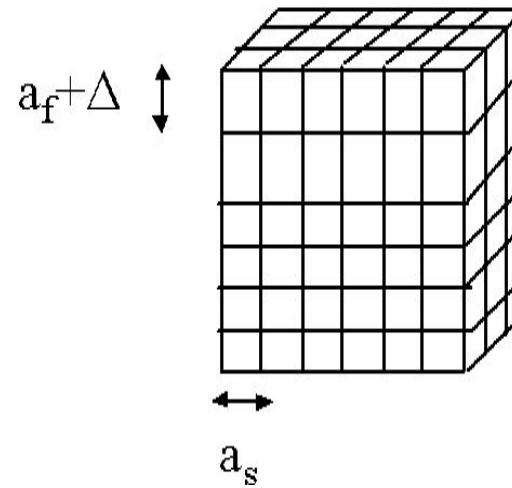
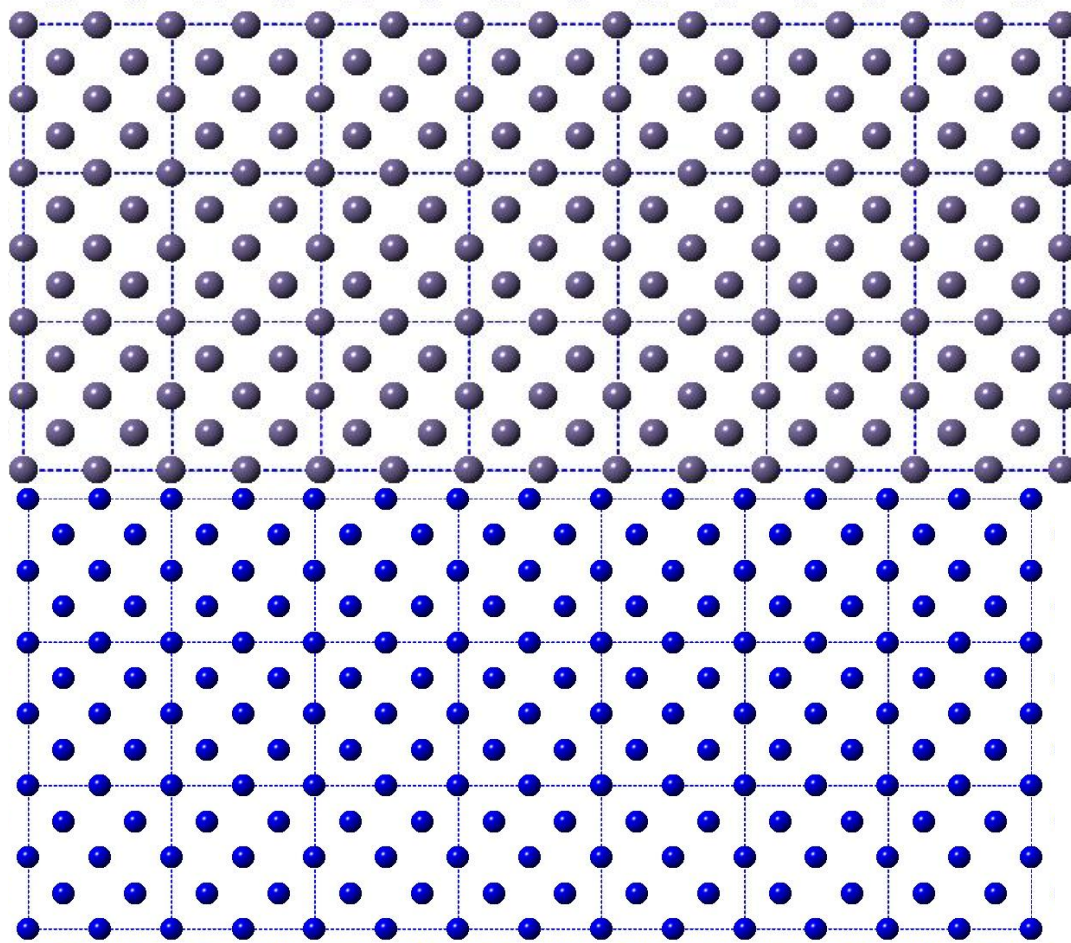
Thickness of SiGe layers

Sharpness of the Si/SiGe interfaces

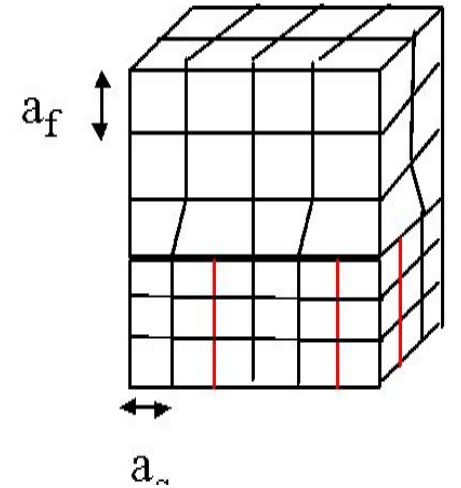
Defects in the multilayer



Misfit Dislocations



Film under strain

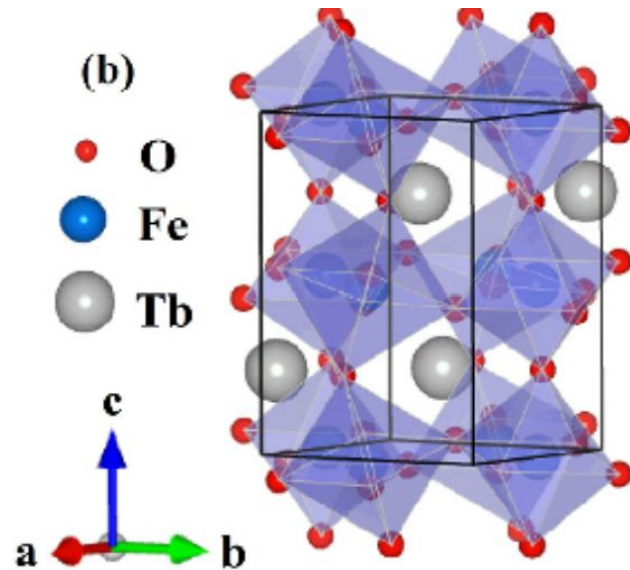


Partially relaxed film

Misfit dislocations (red) form at the interface between film and substrate. Dislocations degrade the properties of the sample.

Sample 2 : TbFeO₃ on YSZ

- Used for photocatalysis, electronics, and magnetic applications.
- TbFeO₃ film deposited by PLD on Y₂O₃ stabilized zirconia Zr₂O₄ with **hexagonal structure**.
- Y_xZr_{1-x}O₃ the value of x determines the crystal structure of YSZ and this controls the crystal structure of TbFeO₃ and its properties



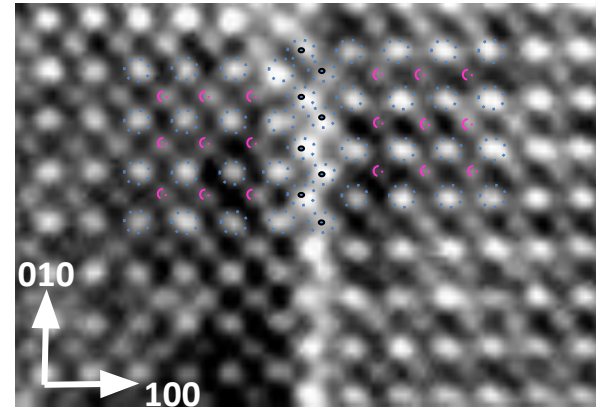
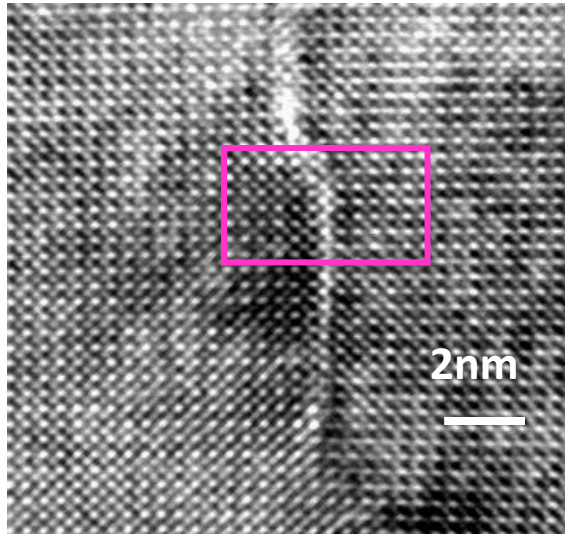
Usual structure Orthorhombic

It is known that under special deposition/synthesis conditions, TbFeO₃ grows in a hexagonal structure, which has a completely different property compared to the orthorhombic one. Using autonomous deposition condition optimization experiment Dr. Takeuchi's team found the best condition to fabricate phase pure metastable hexagonal phase, and this is what we show here.

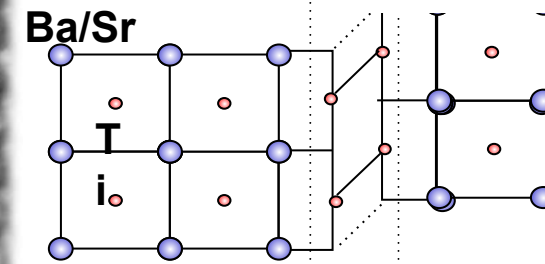
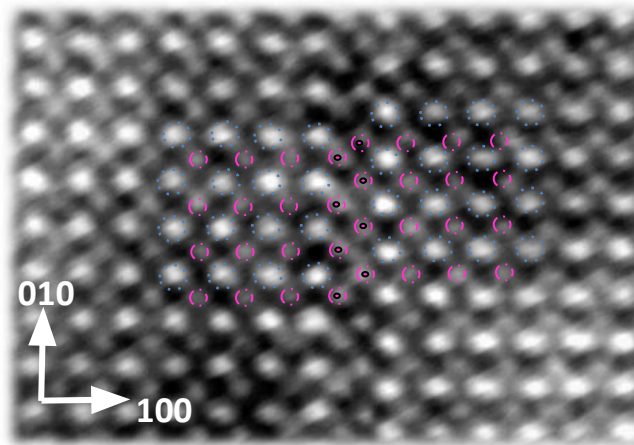
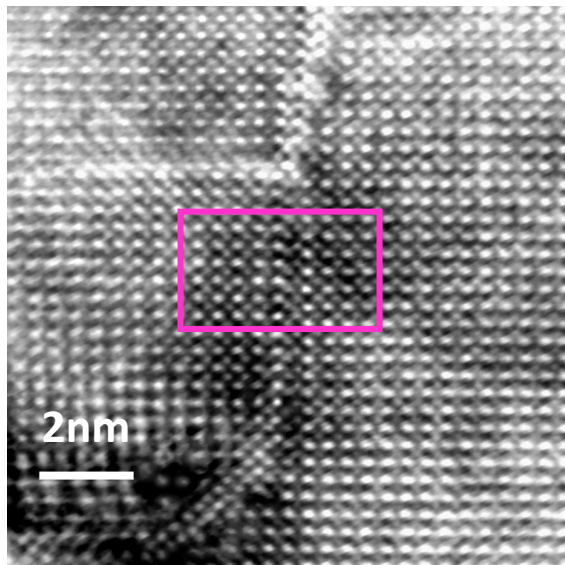
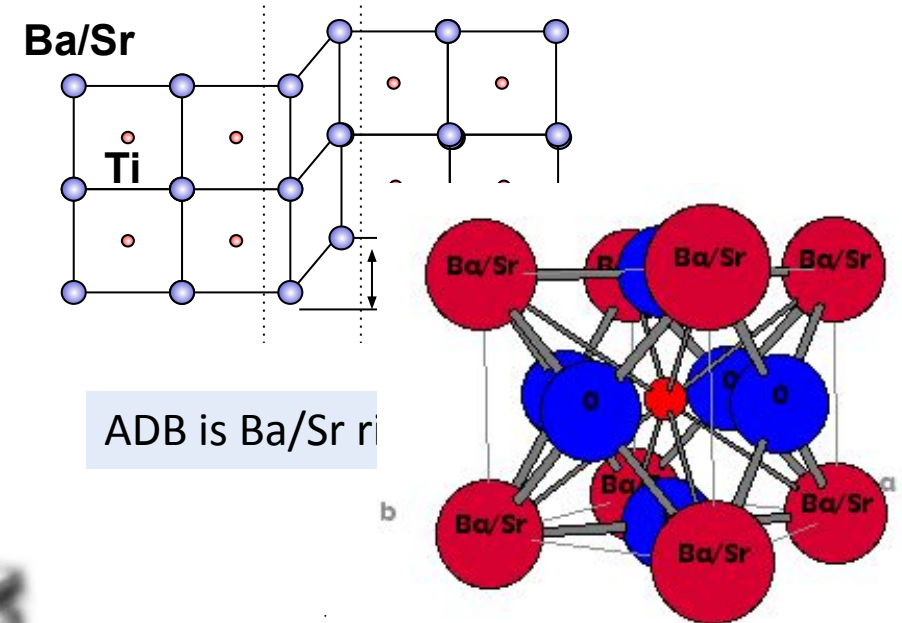
This structure could also have defects that affect the properties.

Cao, Yiming et. Al. (2016). Magnetic phase transition and giant anisotropic magnetic entropy change in TbFeO₃ single crystal. Journal of Applied Physics. 119. 063904. 10.1063/1.4941105.

Antiphase Domain Boundaries In Ba(Sr) TiO₃



● Ba/Sr ● Ti



Section 0.1.2 Histogram and Contrast in Photography

Dynamic range is the ratio between the maximum and minimum measurable light intensities (in photography) or signal amplitudes (in audio) a system can capture, typically measured in stops or decibels (dB). It represents the ability to retain detail in both extreme highlights and deep shadows simultaneously without clipping or noise.¹

A linear detector (or linear image sensor) in electron microscopy is a detector where the signal output is directly proportional to the incident electron flux.

An annular dark field detector (ADF) is a linear detector made of an array of pixels that measure the number of electrons in each pixel to form an image. When a pixel is saturated information is lost because the pixel has reached its maximum charge capacity.

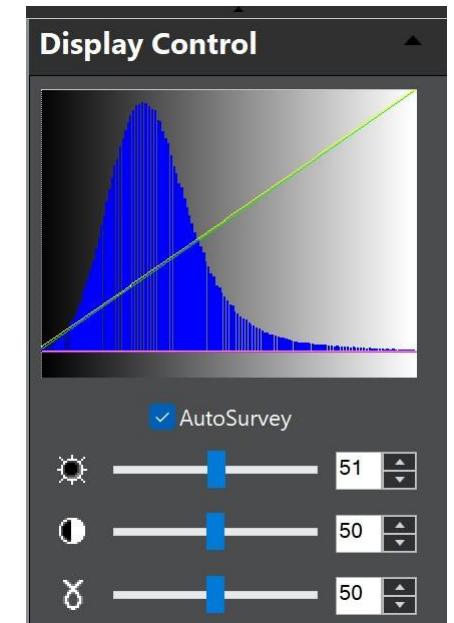
It is important to adjust the dynamical range so that no information is lost.

Contrast is the difference between bright and dark features in an image.

Gamma is a non-linear correction to adjust mid tones without affecting contrast. To

enhance viewing of the image

1. https://en.wikipedia.org/wiki/Dynamic_range



Histogram
Obtained from Gatan Digital
Micrograph



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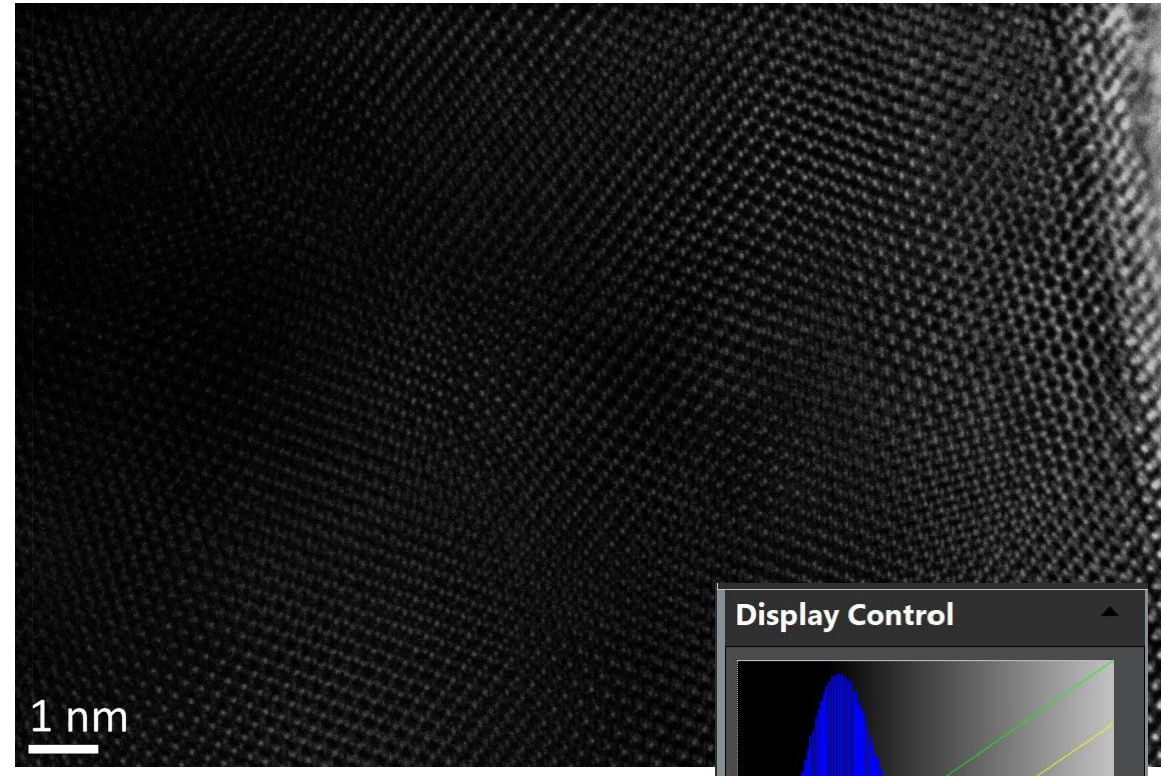
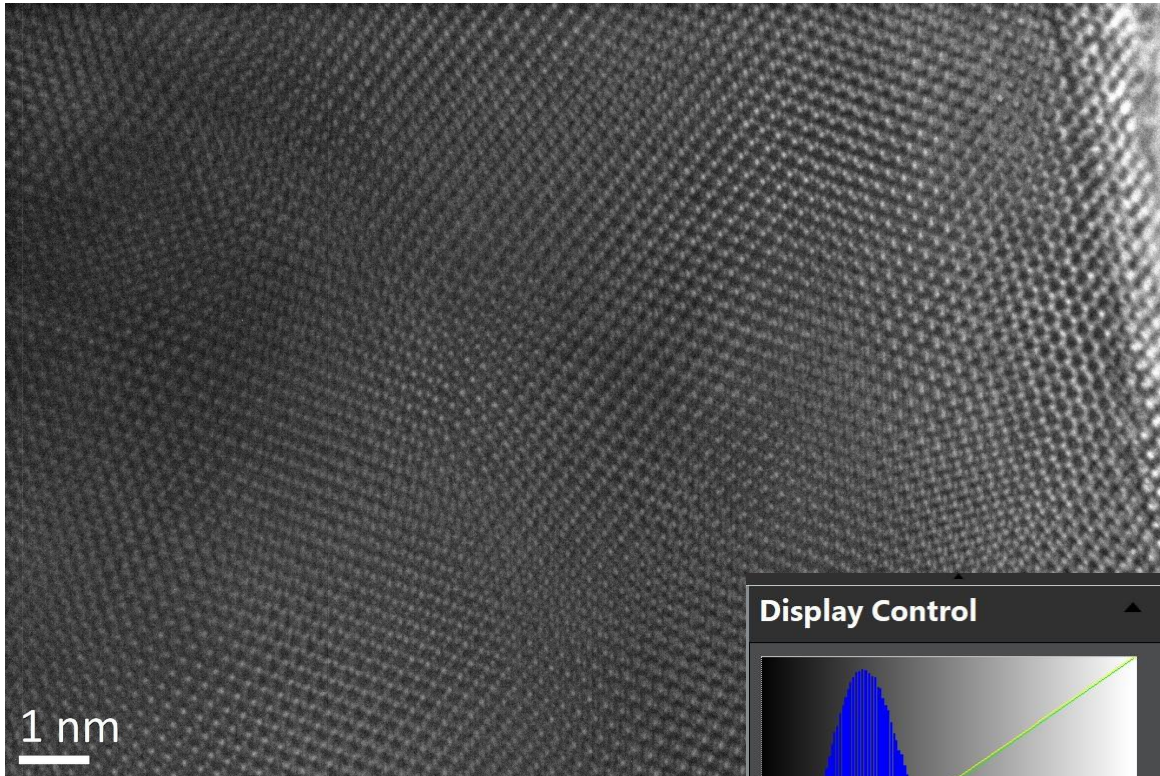


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Effect of brightness



Section 0.1.3 Noise Correction in ~~TEM/STEM~~

Sources of Noise

There can be many sources of noise in TEM/STEM

- Shot noise- from fundamental Physics. Dominant in STEM. It arises from the statistical fluctuations in the number of incident electrons and electron scattering events.
- Detector/Read noise. Variations in the electronics of the detectors in the microscope
- Room temperature variations,
- Variations in the accelerating voltage of the microscope
- Sample modifications (from beam damage or instabilities under the beam, etc)
- Scan jitter (of the e-beam)
- Sample DRIF



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Noise Correction with Gaussian Filter

- Gaussian Filter reduces Gaussian noise by deconvolution of a Gaussian (bell shaped curve) with the image to diminish high frequency noise.
- It is a form of weighted average of pixels around each pixel.
- The center pixel is averaged by 3x3, 5x5, etc pixels (kernel size) with a standard deviation σ .
- This creates smoother transition but if σ is high it can reduce sharpness.

Gaussian noise has a probability density distribution with Gaussian shape. It is usually low.

$$p(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$$

z: grey level, μ mean grey value, and σ standard deviation

Need to correct noise before making measurements to have more accurate values.

Section 0.1.4 DRIFT Correction

In TEM an image is acquired with an exposure time of < 1 sec. During this time the sample could have drifted from original position giving rise to an image which appears to be out of focus.

In STEM an image is acquired as the beam is scanned over an area of the sample with a short dwell time (microsec- millisec) at every spot to minimize drift. This image is not optimized for contrast and brightness.

If a series of images is acquired from the same area they can be summed up to increase intensity and contrast. Prior to combining the images it is necessary to correct for sample drift from one image to the next.



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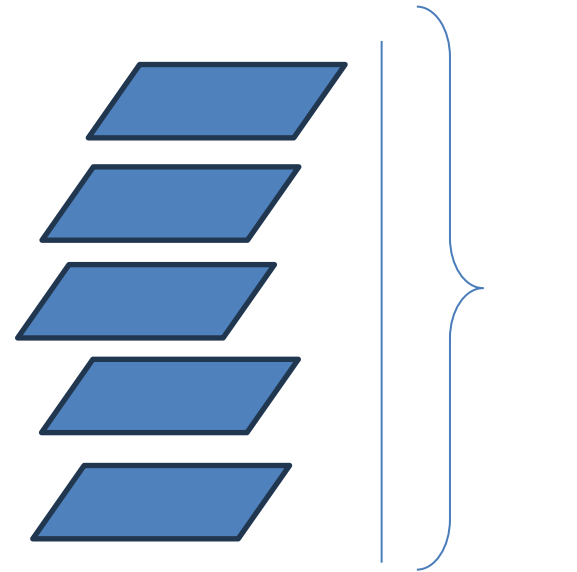
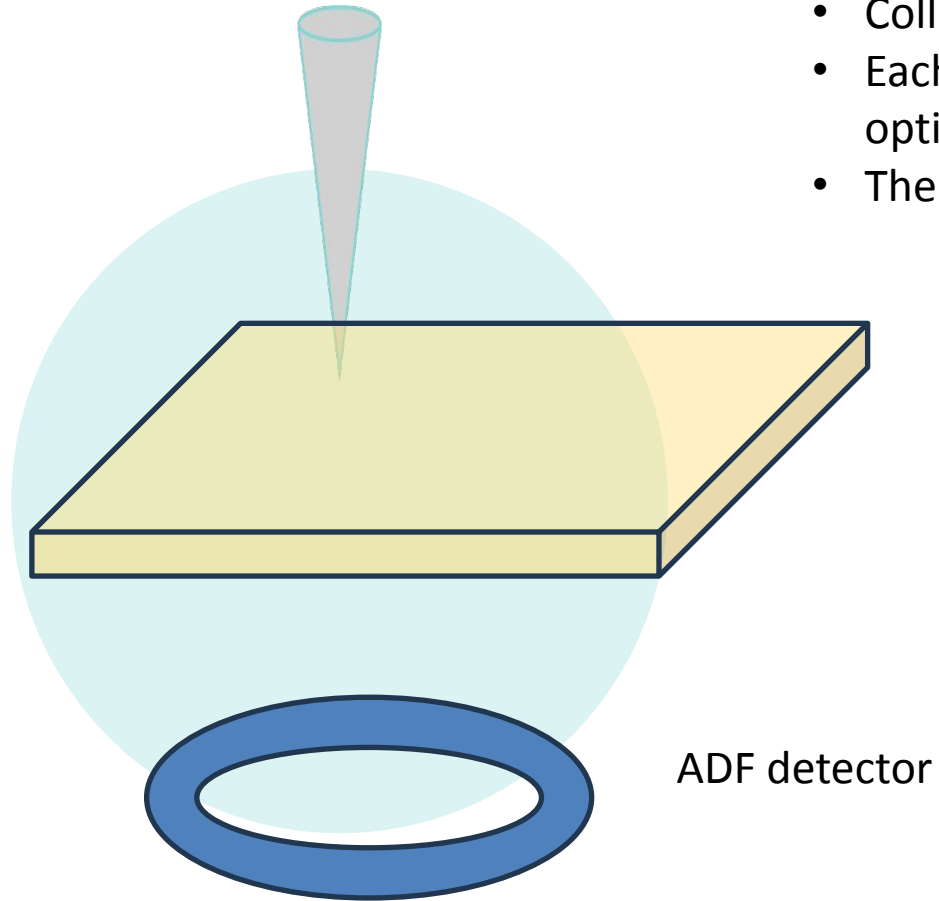
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Drift Correction

- Collection of several images from the same area of the sample.
- Each image is obtained with short dwell time to minimize drift; not optimized for brightness and contrast
- The images are then corrected for drift and summed up



Need to align images

Images are ready to be summed up to optimize brightness and contrast

Section 0.1.5 Measurement of Distances in HRTEM Images

Atomic Resolution Images to measure distances in fraction of nm (pm).

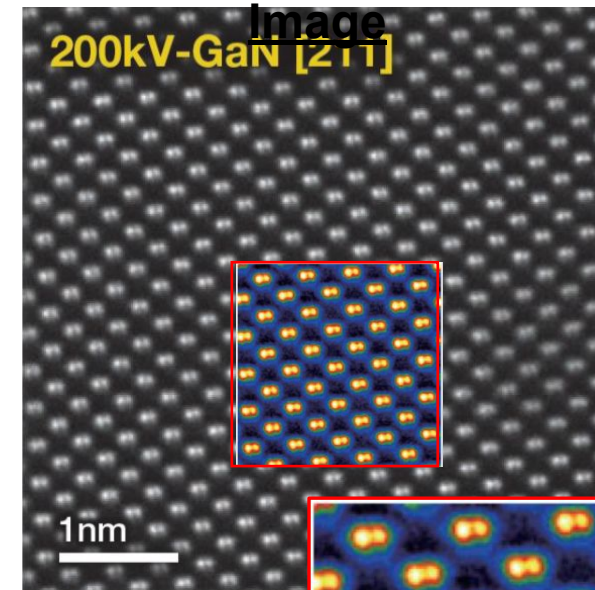
Interplanar Distances.-

Crystalline materials could have changes in their structure due to strain, changes in crystal structure.

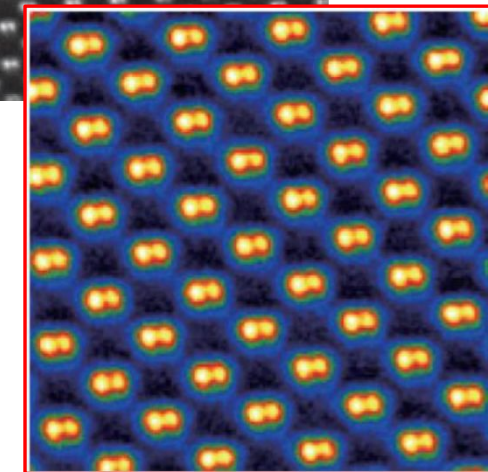
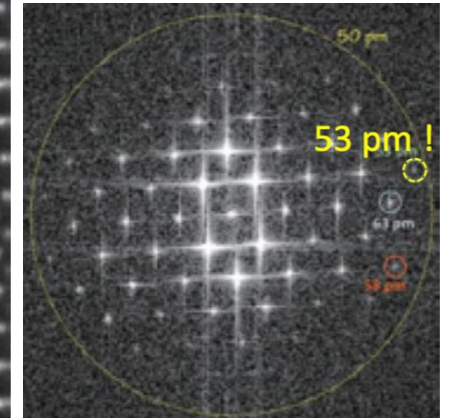
It is important to have a well calibrated microscope to accurately measure distances. The microscope is calibrated using a standard sample, such as STEM images of Si or GaN of known interatomic distances, i.e, distance between Si-Si or Ga-Ga dumbbells

Example: in the $\text{Si}_{20}\text{Ge}_{80}/\text{Si}$ sample the properties of the device depend on the thickness of the Ge, and $\text{Si}_{20}\text{Ge}_{80}$ layers. Variations in the thickness of the Ge or SiGe layers affect the properties.

HAADF STEM



FFT



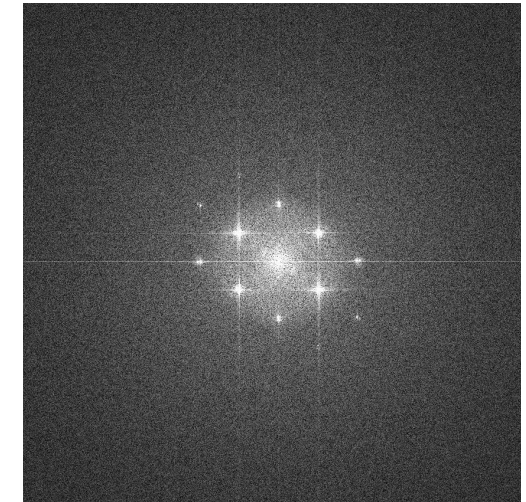
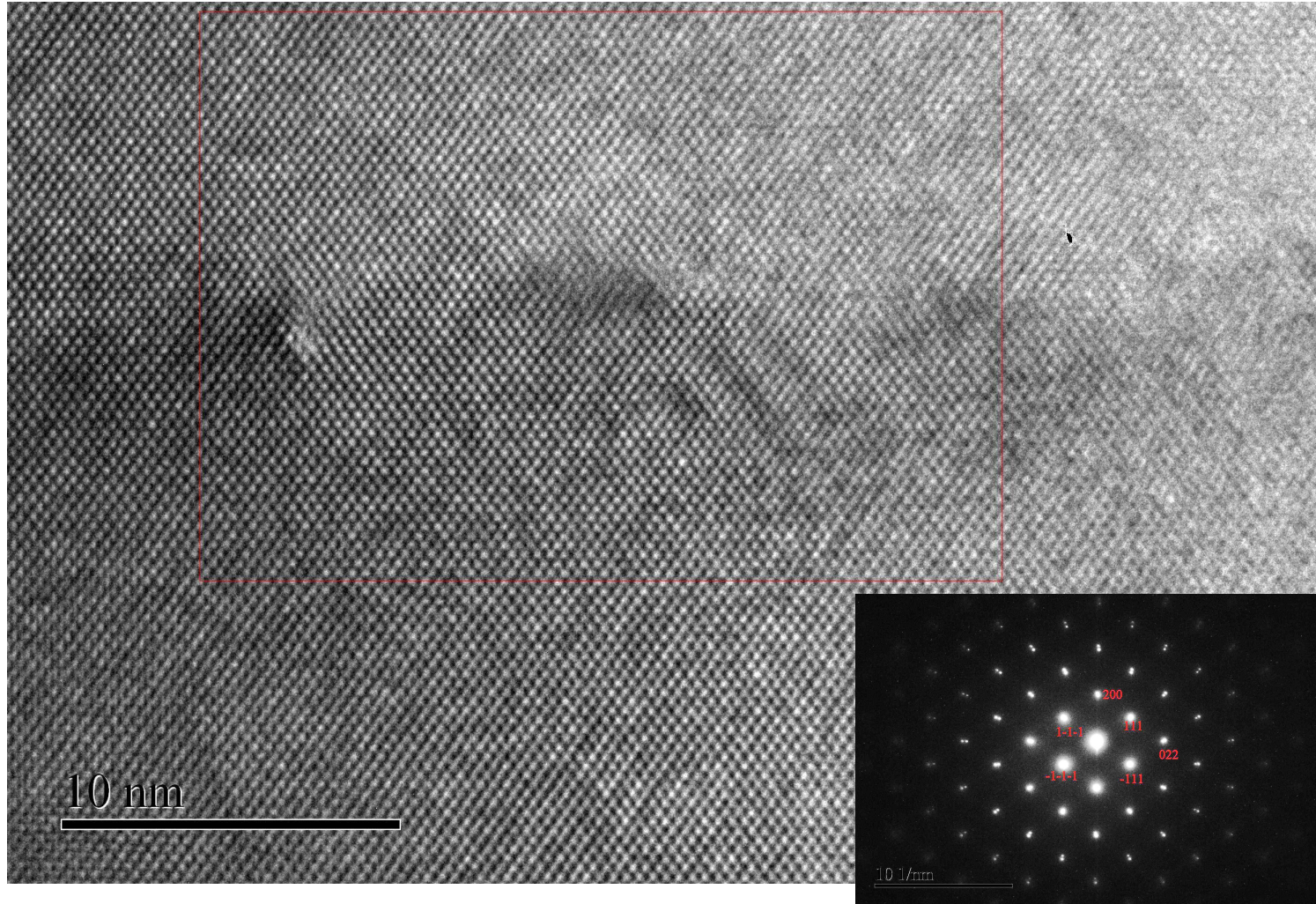
Ga-Ga dumbbells split by 63 pm

From JEOL

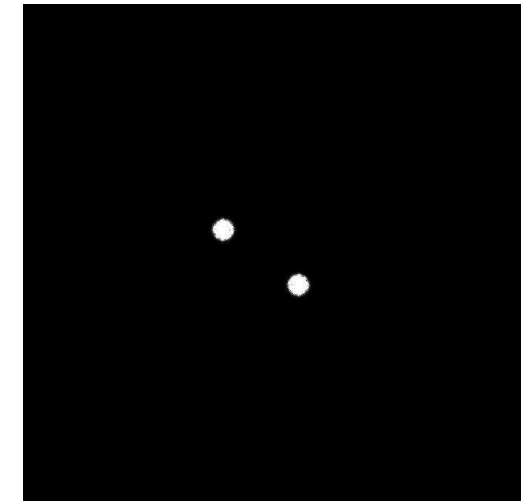
Section 0.1.6 FFT Analysis

InAs

InP

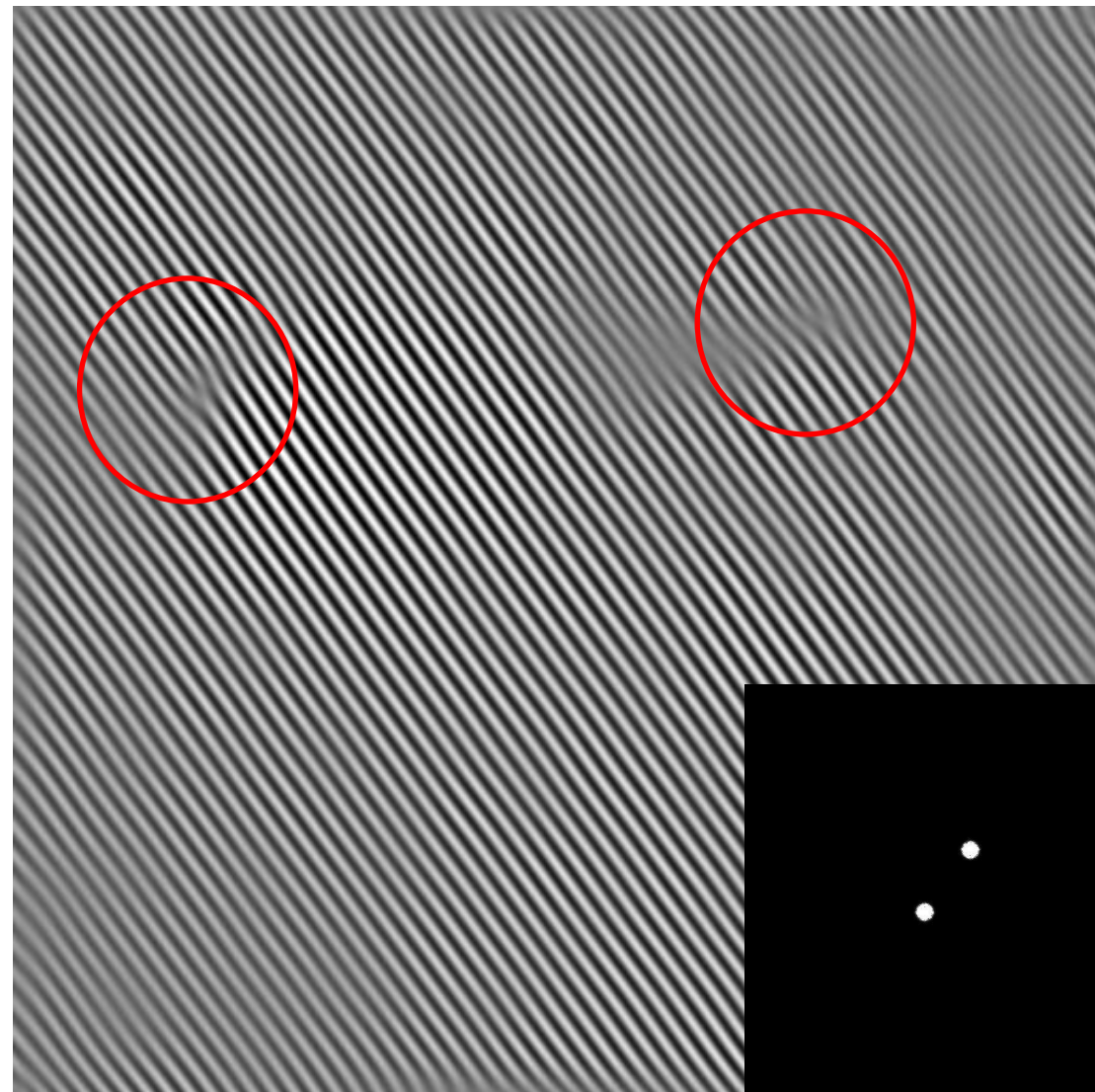
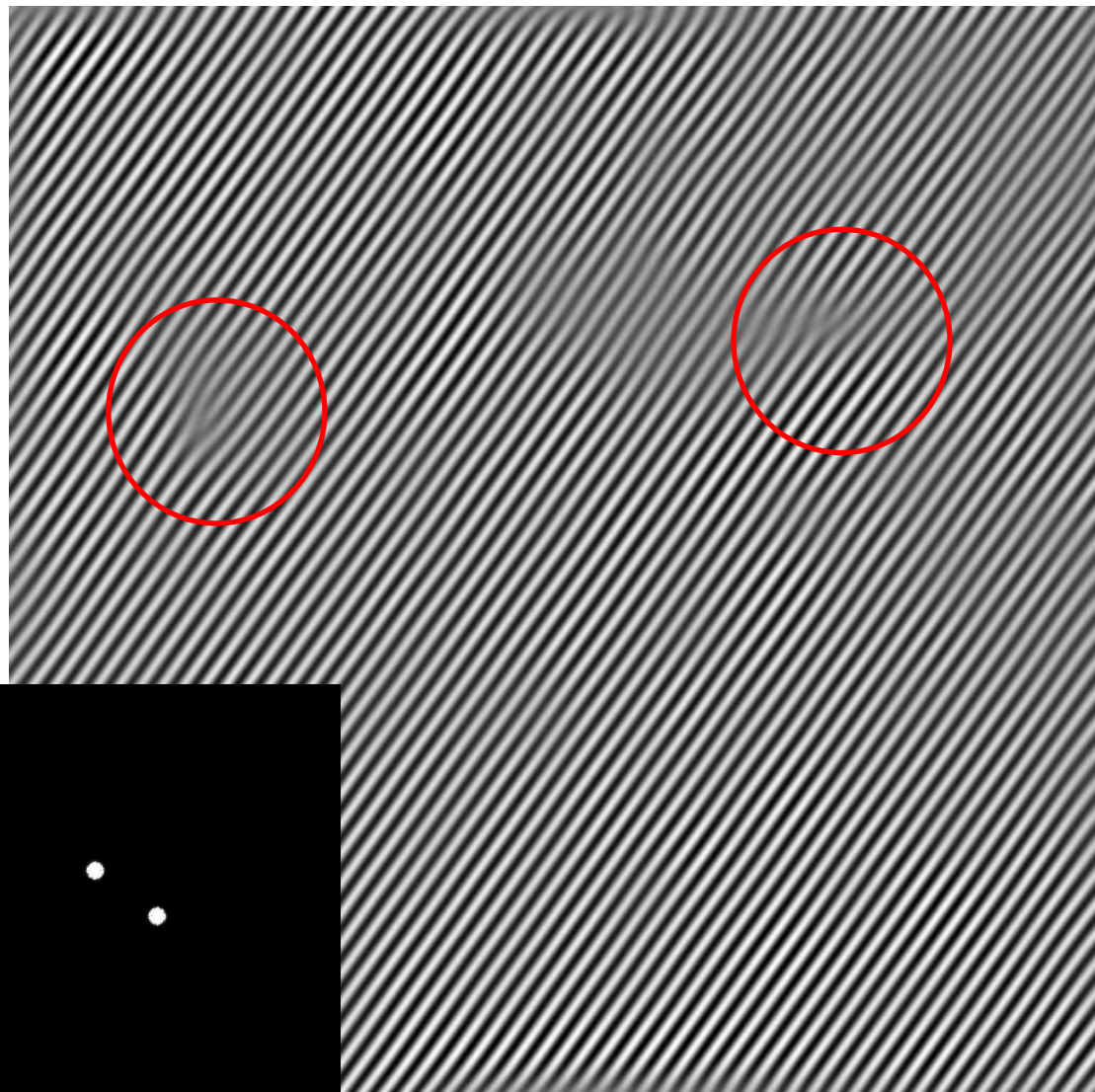


FFT from red box

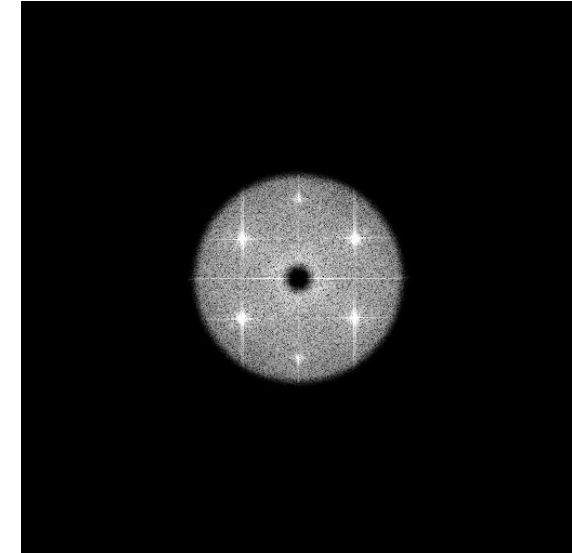
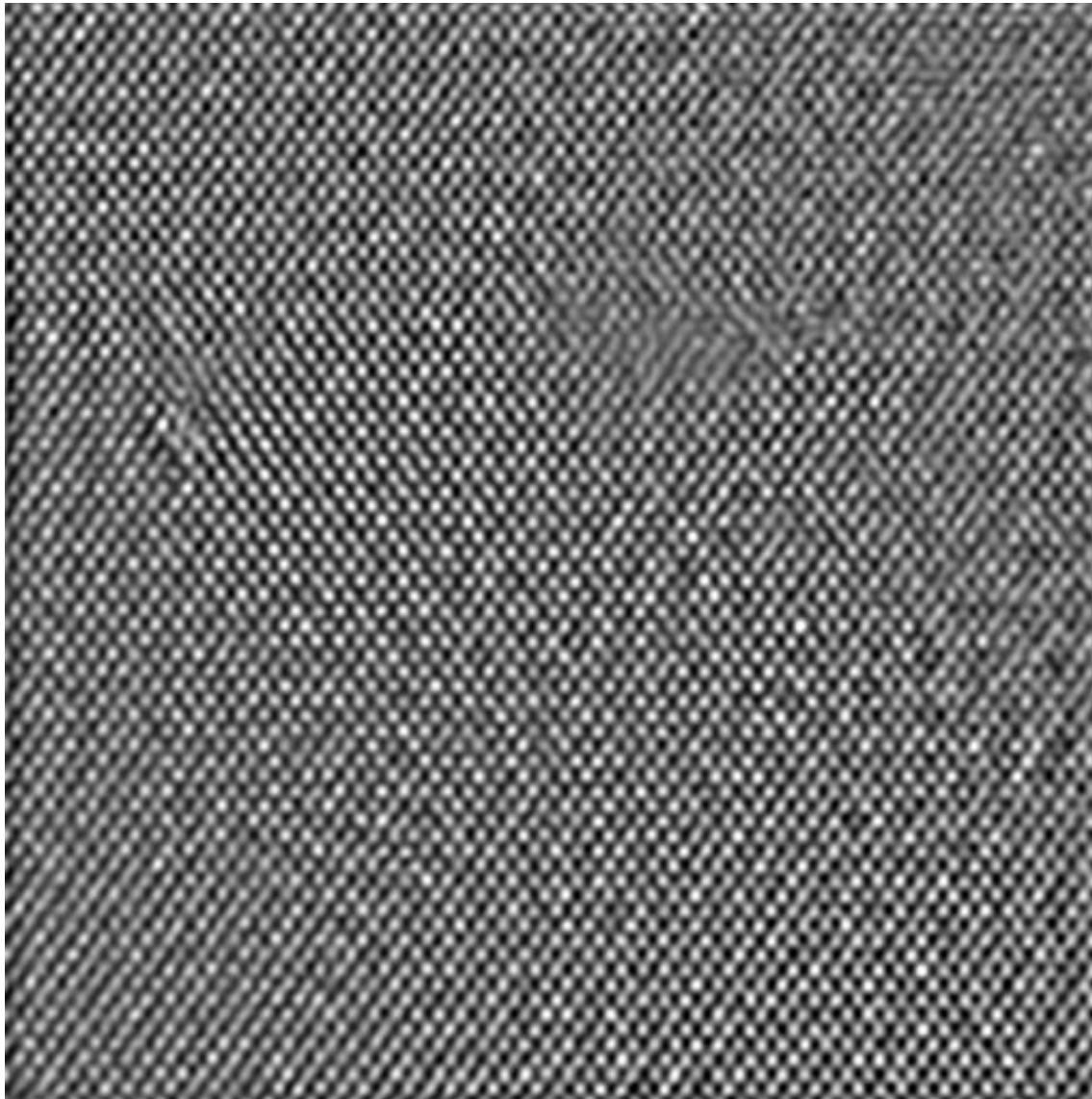


Masked FFT

Inverse FFT (IFFT) Ba(Sr) TiO₃ Misfit Dislocations

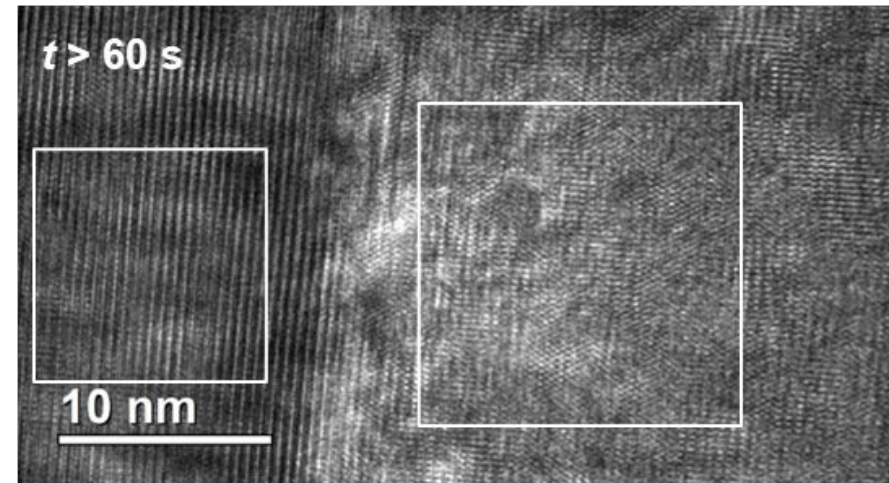
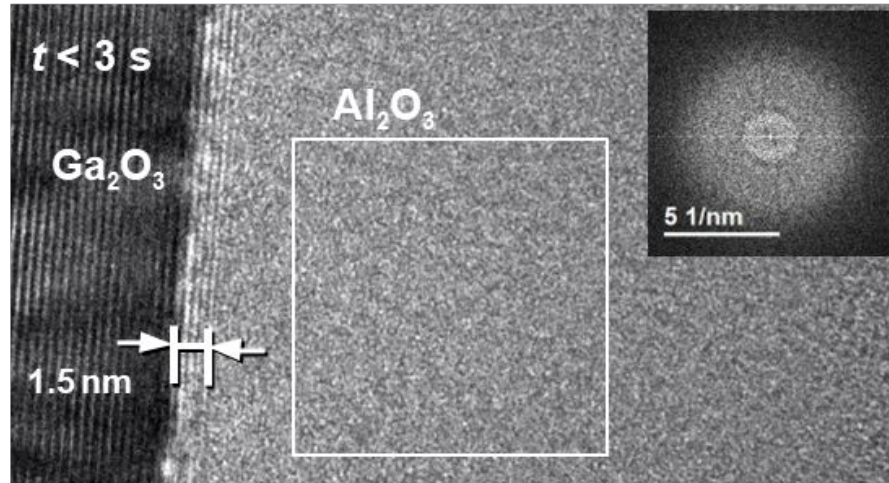


IFFT to Enhance Contrast Filter out noise

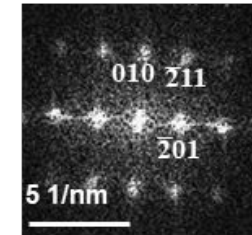


Annular filter

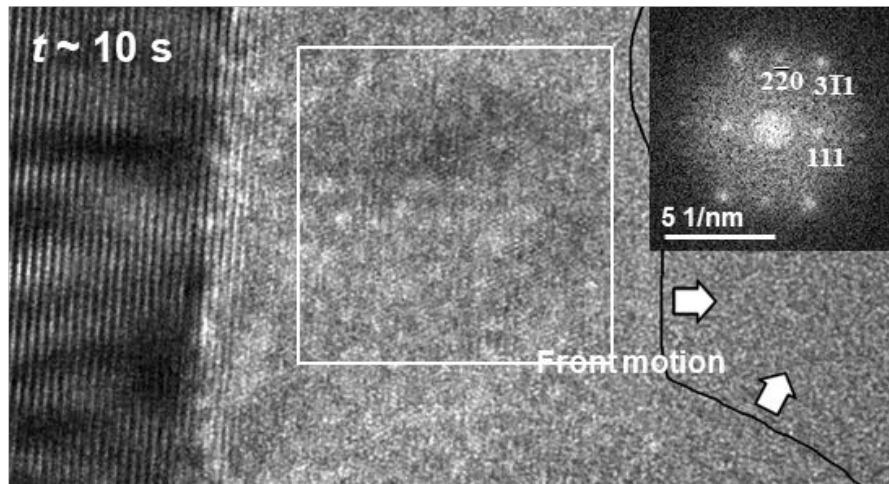
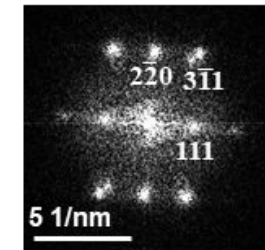
Electron Beam-induced Crystallization $\text{Ga}_2\text{O}_3/\text{Al}_2\text{O}_3$



© [102] Ga_2O_3



© [11 $\bar{2}$] $\gamma\text{-Al}_2\text{O}_3$



- Al_2O_3 layers crystallized fully after several seconds of TEM beam exposure, outward from interface (not from the Al contact, or homogeneously)
- Each of 3 samples showed the same qualitative effect, including one already annealed at 500°C



Thank you for participating in the workshop

Questions?



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